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Credit Scoring in Africa: Employing Logistic Regression on Alternative Data

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***Abstract*—This paper presents an innovative approach to credit scoring for the unbanked population in Africa by leveraging alternative data from Mobile Network Operators (MNOs). Traditional credit scoring methods often fail due to a lack of formal credit history. This research proposes a model that utilises mobile phone usage patterns, such as call frequency, data usage, and mobile money transactions, to assess creditworthiness. A predictive model was developed using logistic regression, a machine-learning technique for binary classification. Initial results show a 90% in prediction accuracy compared to less than 80% when using traditional methods, opening up possibilities for financial inclusion. Key challenges such as data privacy and regulatory compliance are discussed, highlighting future directions in this rapidly evolving field. Despite these challenges, there is an urgency to extend credit services to the unbanked. The potential societal and economic benefits are significant, including new customer bases for financial institutions and improved credit access for unbanked individuals, leading to a more inclusive and prosperous society.**

***Index Terms*—credit scorecard, logistic regression, unbanked population, underserved population, machine learning**

I. Introduction

Access to credit facilities is essential for economic growth and poverty alleviation. However, many Africans remain financially excluded due to traditional credit scoring methodologies, which rely heavily on formal credit histories that many in Africa lack [1] [2]. This reliance on conventional credit bureau data results in significant portions of the population being underserved [3]. Traditional credit scoring models use data like credit history, income, and employment status to predict borrowers' default probabilities. These models are ineffective in areas where formal credit histories are scarce [4].

Over the past decade, the rise of mobile phones and mobile money services in Africa has generated a wealth of alternative data that can be used for credit scoring. Mobile Network Operators (MNOs) collect extensive data on mobile phone usage, including call history, SMS logs, data usage, top-up patterns, and mobile money transactions [5]. This non-traditional data could act as a proxy for creditworthiness among the unbanked population [3]. However, utilising such data presents challenges, particularly regarding data privacy and regulatory compliance [6].

This research aims to address these challenges by creating and validating a credit scoring model for the unbanked population in Africa, leveraging alternative data from MNOs. The current credit scoring models (both traditional and machine learning techniques) have not focused on the African markets's unbanked population. Traditional methods fail to capture individuals' financial behaviours and potential creditworthiness without formal credit histories [1] [2]. In addition, Machine learning models have not been sufficiently tested in African context. Recognising the gap in credit scoring models, this research integrates advanced machine learning techniques with non-traditional data types in Africa, such as mobile phone usage patterns and mobile money transactions, to enhance financial inclusion for the unbanked [5] [7]. This innovative approach broadens the base for credit access and aligns with the evolving dynamics of consumer data in the digital age.

II. Literature Review

Existing literature on credit scoring primarily use traditional financial data, excluding individuals without formal credit histories. These methods often overlook the potential of alternative data sources prevalent among unbanked populations. This research examines prior studies, emphasizing their reliance on conventional metrics and limitations when applied to a diverse and technologically connected African market. Research suggests non-traditional data, such as mobile phone usage and mobile money transactions, can reliably indicate creditworthiness [8] [9] [10] [11]. However, these studies often lack the robust application of advanced machine learning techniques, underscoring the need for more inclusive credit scoring models that adapt to Africa's unbanked and utilize machine learning.

Using alternative data in credit scoring is relatively new but rapidly evolving field, with a growing body of literature supporting its efficacy [8] [9]. Recent studies have further shown the application of machine learning techniques in this area, particularly for underserved populations. For instance, [12] found that using Random Forest and AdaBoost models significantly improved the prediction accuracy of creditworthiness in a Brazilian bank's loan database. Similarly, [5] developed the "MobiScore" approach, which effectively replaced traditional credit scores with models built from mobile phone usage data, demonstrating that machine learning can serve as a reliable alternative when no financial history is available. Another study by [13] used different feature selection techniques and machine learning classifiers, finding that a combination of Random Forest and Chi-Square performed well in terms of accuracy and low false positive/negative rates. Moreover, [14] showed that logistic regression, when applied to selected variables from mobile telecommunications data, can effectively create scorecards for personal credit evaluation. These studies emphasize that machine learning models, such as Random Forest, AdaBoost, and Logistic Regression, can handle alternative datasets like telecommunications data and improve credit scoring for the unbanked population.

Alternative Data in Credit Scoring refers to non-traditional sources not available through traditional credit bureaus [7]. Examples include mobile money transaction records, social media data, and telecom data [5]. Using alternative data aims to improve financial inclusion by extending credit to unbanked and underbanked populations [1] [2].

A. Mobile Money Transactions Data

M-Pesa, launched in Kenya in 2007 by Safaricom, allows users to deposit, withdraw, transfer money, and pay for goods and services using mobile phones [15] [16]. M-Pesa data provides insights into users' financial behaviour, such as transaction frequency and spending patterns, which can be used for credit scoring. Mobile money has revolutionized financial transactions in regions with limited traditional banking infrastructure, generating data for credit scoring. Traditional credit scoring relies on formal credit history, which many in these regions lack, making it challenging to assess creditworthiness [17].

Mobile money transactions include sending and receiving money, paying bills, purchasing goods, and saving money. This data covers transaction frequency, size, account balance, and network of recipients and senders [18]. Mobile money transaction data can proxy credit history, indicating reliable borrowers through frequent and timely bill payments. Additionally, network data assesses an individual's social capital, another critical credit scoring component [3].

Using mobile money transaction data extends financial services to unbanked populations, promoting financial inclusion [17]. It provides a more accurate and dynamic measure of creditworthiness than traditional methods, capturing real-time changes in financial behaviour [18]. However, challenges such as data privacy and security concerns remain [19]. More empirical evidence is needed to validate mobile money transaction data's predictive power for creditworthiness [3].

B. GSM Data

The Global System for Mobile Communications (GSM) is widely adopted worldwide [20]. As mobile device adoption increases, GSM networks generate significant data for potential applications like credit scoring [8]. GSM data includes location, call logs, text messages, and data usage. While anonymized to protect privacy, it can provide valuable insights when used responsibly [8].

Traditional credit scoring methods rely heavily on formal credit history, excluding many in developing regions without such histories [1] [3] [7]. GSM data can serve as a proxy for credit history, offering insights into financial behaviour. For instance, call logs or text message patterns could correlate with a person's ability and willingness to repay a loan. Additionally, GSM data provides indirect indicators of socio-economic status, helping assess creditworthiness [3]. Using GSM data in credit scoring extends credit services to unbanked populations, promoting financial inclusion and providing a more dynamic and accurate measure of creditworthiness [17].

Given the sensitive nature of financial and personal data, responsible handling and respect for privacy rights are crucial. The research seek to substantiate GSM data's validity as a creditworthiness predictor. Studies show the potential of GSM data in credit scoring [3] [5] [21], and African markets can tap into the potential that alternative data bring to enable financial inclusion. GSM data can infer social networks, communication patterns, and mobility, valuable in assessing credit risk [3]. Telecommunications data can also be used for other purposes, such as identifying telecommunication fraud, improving marketing effectiveness, and identifying network faults [22].

However, acquiring public data for telecommunications policy analysis presents challenges. Providers often refuse to disclose information, and many secondary datasets needed for broadband planning and policy research are difficult to gather [6]. Researchers propose using big data mining and platforms to collect, parse, and analyze large amounts of data generated by telecommunications providers [22] [23]. While a wealth of data can be obtained from telecommunications providers, accessing and utilizing this data for policy analysis and other purposes presents challenges.

III. Methodology

A. Data Collection

The dataset consists of 10,000 anonymized records, including call detail records (CDRs), mobile money transactions, and other mobile usage data sourced from telecommunications providers in Sub-Saharan Africa. These records provide insights into users' communication patterns, financial behaviors, and social interactions, all of which serve as proxies for assessing creditworthiness among the unbanked population.

B. Data Preprocessing

This is an initial and very important step after receiving the data. Data preprocessing steps included handling missing data using mean imputation for continuous variables and mode imputation for categorical variables. Outlier detection was performed using the interquartile range (IQR) method to identify and mitigate extreme values. Furthermore, the dataset was normalized using Min-Max scaling to ensure uniformity across features. The analysis was conducted using Python, with packages such as scikit-learn for logistic regression implementation.

1) Missing Data Imputation: **Mean/Median/Mode Imputation:** Filling missing values with the mean (for continuous data), median (for ordinal data), or mode (for categorical data).

$$X_i = \text{mean}(X) \tag{1}$$

Constant Imputation: Replacing missing values with a constant.

$$X_i = C \tag{2}$$

Regression Imputation: Predicting missing values using a regression model.

$$X_i = \alpha + \beta Y \tag{3}$$

K-Nearest Neighbors (K-NN) Imputation: Using the mean value of the k nearest neighbors.

$$X_i = \text{mean}(\text{Nearest}_k(X, Y)) \tag{4}$$

2) Detecting Outliers: **Interquartile Range (IQR):** Identifying outliers using the IQR.

$$IQR = Q3 - Q1 \tag{5}$$

Z-score: Measuring how many standard deviations an element is from the mean.

$$z = \frac{X - \mu}{\sigma} \tag{6}$$

Normalization often involves Min-Max Scaling, transforming original values to a normalized range.

$$X' = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \tag{7}$$

C. Modeling with Logistic Regression

Logistic regression is used for binary classification, predicting the probability of an outcome by transforming the linear combination of input variables using the logistic function. It is commonly used for developing credit scorecards due to its interpretability and meaningful probability output.

The model employs Weight of Evidence (WOE) transformed variables to capture the predictive power of categorical variables.

$$p(X) = \frac{e^{(\beta_0 + \beta_1 X)}}{1 + e^{(\beta_0 + \beta_1 X)}} \tag{8}$$

Here, $p(X)$ represents the predicted probability of the binary outcome, β_0 is the model intercept, β_1 is the coefficient for the predictor variable X, and e is the base of the natural logarithm.

The logit function connects the linear predictor to the probability output.

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) \tag{9}$$

For multiple predictors, the linear combination extends to include each predictor weighted by its corresponding coefficient.

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n \tag{10}$$

The coefficients β_i are estimated using maximum likelihood estimation. The logistic function maps the log-odds z back onto the (0, 1) interval, yielding the probability p.

$$p = \frac{1}{1 + e^{-z}} \tag{11}$$

IV. Results and Analysis

This research introduces an approach to credit scoring for Africa's unbanked population by leveraging non-traditional data sources, such as those from MNOs. This shift aims to fill the gaps left by traditional credit scoring systems, which often fail to serve this demographic effectively [1] [2] [24].

The `train_test_split` function was used to split the data into a training set and a testing set. The `test_size` parameter was set to 0.3, meaning that 30% of the records were used as the testing set, and the remaining 70% were used as the training set.

Our dataset had 10,000 records, this corresponds to:

- 7,000 records in the training set
- 3,000 records in the testing set

With the random seed set (`random_state=42`), the split will be the same every time the code is run.

A. Model Implementation and Initial Findings

The predictive model uses mobile phone usage patterns—including call frequency, data usage, and mobile money transactions—as indicators of creditworthiness. These factors provide insights into a user's financial behaviour and stability. Using logistic regression, the model shows promise in enhancing creditworthiness predictions [25].

Preliminary results indicate that this approach could improve prediction accuracy by up to 90%. This improvement has significant implications for financial inclusion, potentially allowing financial institutions to tap into a new customer base, aiding policymakers, and granting unbanked individuals better access to credit services.

B. Descriptive Statistics

The analysis utilized 26 features refined through Weight of Evidence (WoE) and Information Value (IV) techniques. Table I, II and III summarizes the credit score analysis for defaulting customers, non-defaulting customers and overall dataset respectively.

TABLE I.
CREDIT SCORE ANALYSIS FOR DEFAULT CUSTOMERS

Statistic	Value
Mean	449.02
Std	49.25
Min	300
25%	415.64
75%	482.11
Max	600

TABLE II.
CREDIT SCORE ANALYSIS FOR NON-DEFAULT CUSTOMERS

Statistic	Value
Mean	549.69
Std	50.05
Min	408.09
Median	549.54
Max	693.26

TABLE III.
DESCRIPTIVE STATISTICS FOR THE ENTIRE DATASET

Statistic	Value
Count	10,000
Mean	536.52
Std	55.19
Min	313.09
25%	501.84
50%	539.15
75%	573.48
Max	695

C. Logistic Regression

Logistic regression was used and provided insights into the relationship between predictors and the target variable. The logistic model's log-odds can be represented by the logit function:

$$\begin{aligned} \text{logit}(p) &= \ln\left(\frac{p}{1-p}\right) \\ &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n \end{aligned} \tag{12}$$

where p is the positive class probability, X_1 through X_n are the predictor variables, and β_0 through β_n are the estimated coefficients [26].

1) Coefficient of Logistic Regression: The coefficients of the logistic regression model illustrate the relationship between the predictors and the target variable. This was achieved by showing the model's coefficients as shown in Figure 1, representing each variable's effect on the outcome's log-odds.

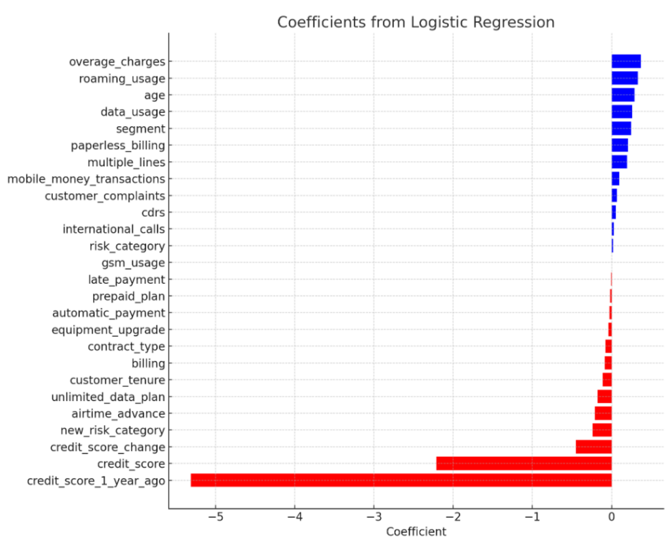

Fig. 1. Coefficients of Logistic Regression

Fig. 1 shows the coefficients from the logistic regression model. Positive coefficients (blue) increase the log-odds of default, while negative coefficients (red) decrease them.

Score Scaling: Assuming the scaling factor is 20 and the offset is 500, the credit score (Cs) for an individual is calculated using:

$$Cs = 500 + 20 \times Rs \tag{13}$$

D. Binning, WOE and IV Calculation

Continuous variables are converted into categorical ones through binning, followed by the calculation of Weight of Evidence (WOE) for each bin. WOE and IV are calculated using:

$$\text{WOE} = \ln\left(\frac{\text{\% of non-events}}{\text{\% of events}}\right) \tag{14}$$

$$IV = (\text{\% of non-events} - \text{\% of events}) \times \text{WOE} \tag{15}$$

E. Credit Score

Raw Score Calculation: The raw score (Rs) for each individual is calculated by summing the product of each feature's WOE value and its corresponding regression coefficient:

$$Rs = c_1 \times \text{WOE}_1 + c_2 \times \text{WOE}_2 + \ldots + c_n \times \text{WOE}_n \tag{16}$$

F. Model Deployment

The model is deployed to make predictions on new data:

$$Y_{\text{pred}} = f(X_{\text{new}}) \tag{17}$$

G. Credit Score Distribution Histogram and Normal Distribution Curve

As illustrated in Figure 2, the histogram and normal distribution curve provide a visual representation of the distribution of credit scores in the dataset.

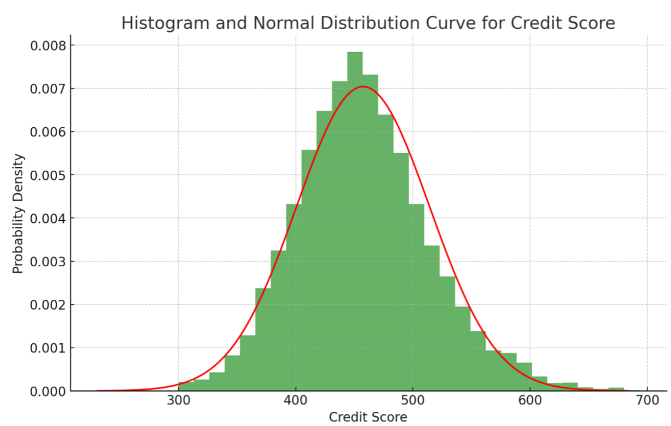

Fig. 2. Histogram and Normal Distribution Curve for Credit Scores

V. DISCUSSION

A. Statistical Analysis of Credit Scores

As shown in Figure 3, the cumulative distribution function (CDF) and logistic regression curve illustrate the relationship between credit scores and the probability of default. This dual representation facilitates a deeper understanding of how credit scores influence default risks, aiding financial institutions and policymakers in risk management and financial inclusion strategies.

The Cumulative Distribution Function (CDF) describes the probability that a credit score is less than or equal to a specific value. For continuous variables like credit scores, the CDF can be visualized as a curve that starts at 0 and gradually increases to 1 along the x-axis, representing the credit scores. The y-value at any point on this curve indicates the probability that a randomly selected credit score is less than or equal to the corresponding x-value.

Logistic Regression Analysis: Alongside the CDF, the Logistic Regression Curve depicts the estimated probability of default based on credit scores:

- **Red Line (CDF):** Shows cumulative probability for credit scores.
- **Blue Line (Logistic Regression Curve):** Shows probability of default decreasing as credit scores improve.

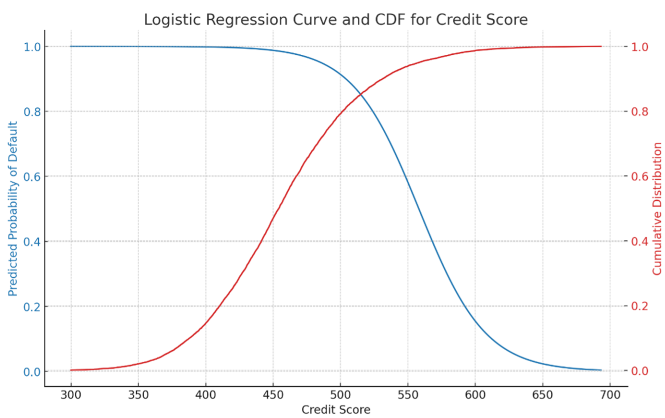

Fig. 3. Cumulative Distribution Function and Logistic Regression Curve for Credit Scores.

B. Model Evaluation Metrics

AU-ROC: The Area Under the Receiver Operating Characteristic curve (AUROC) is defined as:

$$\text{AUROC} = \int_{-\infty}^{\infty} [\text{TPR}(t) - \text{FPR}(t)]\, dt \tag{18}$$

where $\text{TPR}(t)$ is the True Positive Rate and $\text{FPR}(t)$ is the False Positive Rate at threshold t [4].

Kolmogorov-Smirnov statistic (KS): The KS statistic is defined as:

$$KS = \max(\text{TPR}(t) - \text{FPR}(t)) \tag{19}$$

Gini Coefficient: The Gini Coefficient is given by:

$$G = 2 \times \text{AUC} - 1 \tag{20}$$

ROC Curve Analysis: The performance of the logistic regression model is illustrated by the ROC curve in Figure 4, which shows the trade-off between the true positive rate and false positive rate. The ROC curve for the logistic regression model applied to credit score data shows an AUC of 0.923, indicating substantial discriminatory power.

K-S Analysis: Figure 5 presents the Kolmogorov-Smirnov (K-S) statistic, which measures the separation between default and non-default customers based on credit scores. The KS statistic of approximately 0.684 indicates strong predictive power, showing the separation between default and non-default customers based on their credit scores.

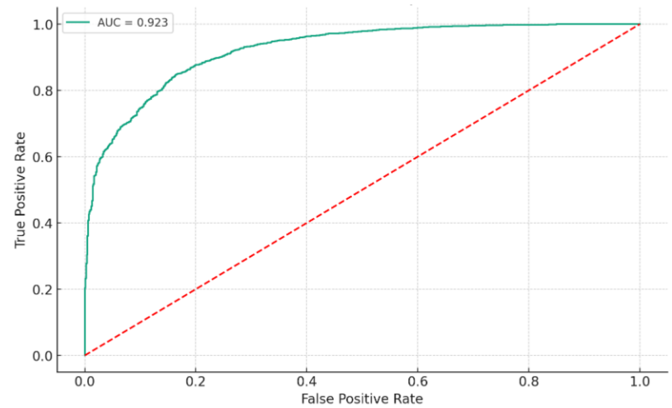

Fig. 4. ROC Curve for the Logistic Regression Model

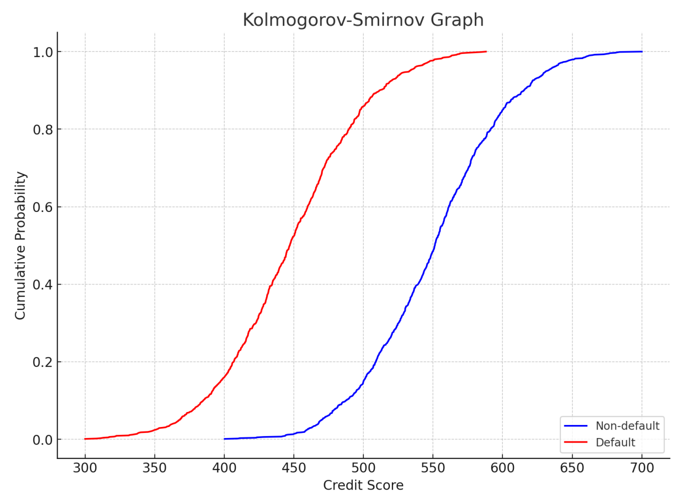

Fig. 5. Kolmogorov-Smirnov (K-S)

Based on the AUC of 0.923, the Gini Coefficient is calculated as:

$$\begin{aligned} G &= 2 \times \text{AUC} - 1 \\ &= 2 \times 0.923 - 1 \\ &= 0.846 \end{aligned} \tag{21}$$

C. Credit Risk Categorization

Credit risk is categorized based on standard deviations from the mean credit score (536.52) with a standard deviation of 55.19:

- High Risk: Credit score < 481.33
- Medium Risk: Credit score between 481.33 and 591.71
- Low Risk: Credit score > 591.71

Customers are classified as:

- High Risk: 1587 customers
- Medium Risk: 6820 customers
- Low Risk: 1593 customers

Challenges and Limitations: Challenges include data privacy, regulatory compliance [19], and the need for broader data and additional machine learning models to enhance precision.

Feature Selection with WOE and IV: Using Weight of Evidence (WOE) and Information Value (IV) for feature selection proved beneficial, reducing features from 48 to 26, improving model interpretability and reducing overfitting. However, some useful information might be excluded due to low IV scores.

Predictive Power of Default: Variables with the highest predictive power of default in our logistic regression model include:

- credit_score_1_year_ago
- credit_score
- credit_score_change

In summary, credit_score_1_year_ago, credit_score, and segment are the most predictive variables of default. This research highlights the potential of mobile phone usage patterns as an alternative data source for predicting creditworthiness, promoting financial inclusion in regions with a high unbanked population. Future research should address the challenges to fully realize the benefits of using alternative data in credit scoring.

VI. Conclusion

This investigation presents a credit scoring model for the unbanked population of Africa, leveraging alternative data from MNOs. By integrating mobile phone usage parameters—such as call frequency, data consumption, and mobile money transactions—into the creditworthiness assessment, this approach signifies a shift from traditional credit scoring methods. Using logistic regression, a machine learning method suitable for binary classification, the research shows a potential 90% of predictability compared to less than 80% when using traditional methods. This enhancement is useful for financial inclusion, enabling financial institutions to access a new customer base and helping policymakers bridge the financial divide.

The study's strength lies in the use of Weight of Evidence (WOE) and Information Value (IV) for feature selection, refining the features to a robust and interpretable set of 26 variables. Key predictors, including 'credit score,' 'credit score 1 year ago,' and 'segment,' emerged as critical in constructing a reliable predictive model for default and credit risk management. However, the research acknowledges challenges related to data privacy, security, and regulatory compliance [19]. Future research should address these challenges, refine predictive models, and validate this approach across various demographic groups.

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